Cyclistic New York Bike-Share Demand Executive Summary

Overview

This project concentrates on investigating Cyclistic customers' bike usage in New York and factors affecting it

The Problem

Cyclistic's Customer Growth Team is working on the next year business plan for New York city. The main task standing before them is growing Cyclistic customer base. For this purpose they want to know:

- How does the demand on different bike stations look like?
- How do different types of riders use Cyclistic bikes?
- How weather and time conditions affect bike usage?

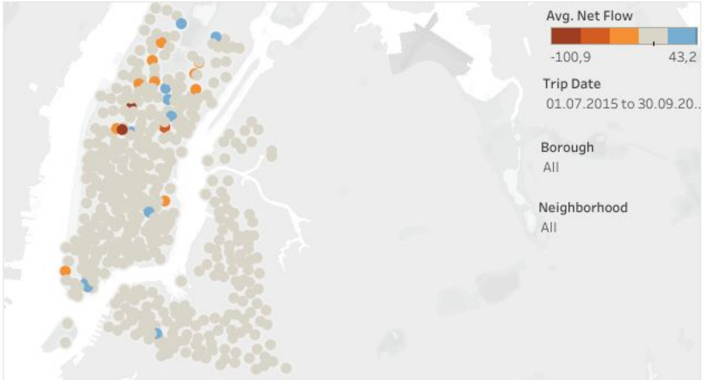
The Solution

The Cyclistic NY has created a data team to analyze data collected by Cyclistic system and create data visualizations that help answer questions posed

Details

Keys insights

- Start trip counts are unevenly distributed: the peak usage is observed in Manhattan (West and Lower East sides), the minimum usage is recorded in Queens and Brooklyn.
- In the context of year peak usage time is recorded from May to October. The reason for the sharp year-on-year growth in November and December of 2015 is determined by low base effect. The average monthly temperature in November and December of 2014 is noticeable lower than a year later.
- There is not a significant shift in net flow number at most stations. While some stations in West, Upper and Lower East Manhattan have negative net flow. However in each region there are stations with positive net flow that we can use to balance.
- The trip count is low than average when temperature is lower than 40° and higher than 80° Fahrenheit, wind speed is higher than 6,5 mph, precipitation is observed and from 21:00 at the evening till 7:00 at the morning. During this time, maintenance work can be carried out
- The share of customers increases on weekends from average 8.5% to 20%. While the number of subscribers' rides decreases on weekends. Because subscribers more actively use bikes for commuting to and from work



Congestion Level at Start Stations

Station Name	Neighborhood	Borough	
1 Ave & E 15 St	Lower East Side	Manhattan	1,8
1 Ave & E 18 St	Lower East Side	Manhattan	1,2
1 Ave & E 30 St	Gramercy Park and Murray Hill	Manhattan	2,6
1 Ave & E 44 St	Gramercy Park and Murray Hill	Manhattan	3,5
1 Ave & E 62 St	Upper East Side	Manhattan	-8,6
1 Ave & E 68 St	Upper East Side	Manhattan	-13,2
1 Ave & E 78 St	Upper East Side	Manhattan	7,8

Reflections/ Next Steps

- Agree on the final dashboard.
- Demonstrate and, if necessary, teach how to work with filter systems on maps.
- Develop ways to better determine the level of congestion at stations to make more efficient decisions.